LabTracer Protocol
User's Manual

Manual Version 2.0
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physics and chemistry

1. LabTracer protocol summary

LabTracer is Windows operating remote program on computer (IBM type PC) to controlling and monitoring for physics and chemistry laboratory test equipment. Using of Lab Tracer offered for user to convenient operating the Jeiotech' physics and chemistry laboratory test equipments which is included communication function

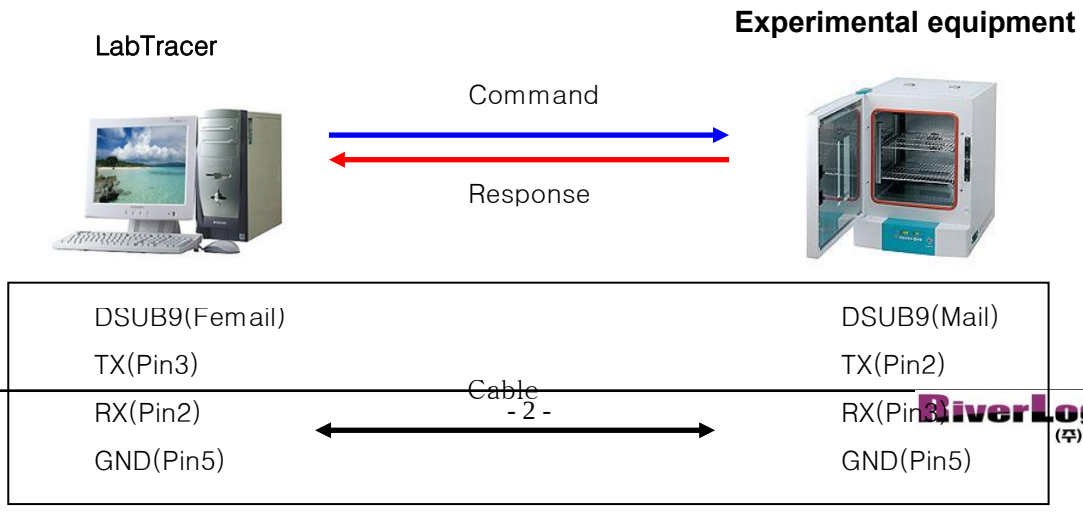
Using Lab Tracer, the user can indicate test equipment information (Set temperature, Temperature sensor output, quantity of heating, equipment condition) on a computer, even the equipments isn't located easy connecting environments. Also user can be understood the equipment conditions easily and conveniently by graph style data. (Not Text style data). This document provided protocol for equipment operator, if operators want to make or use similar and unique programs with Lab Tracer.

This protocol is communication protocol between experimental equipment and operating program (Substitute for Lab Tracer). Use of this protocol, the equipment user can be operating with their own program of Jeiotech physics and chemistry laboratory test equipments on the computer. Also the program developer can make their unique and convenient program by our provide protocol. Lab Tracer will be upgrading user' convenience, and we will support user to make and develop their new programs by using protocol.

In this document, the word of LabTracer when we explain the protocol. The Lab Tracer mean is same as with user have developed unique program.

2. Communication connecting

For the Lab Tracer software operate properly, the unit must be connected to your computer using RS-232 communications cable supplied. Communication settings is 19200bps, 8bit, 1Stop bit, None Parity. Using RS-232 cable' connection terminal part with computer is DTE(DSUB9 Femail) type and test equipment connection terminal part is DCE(DSUB9 Mail) type. The computer part cable' Terminal Pin 2, 3, 5 will connect each equipment terminal Pin2, 3, 5.



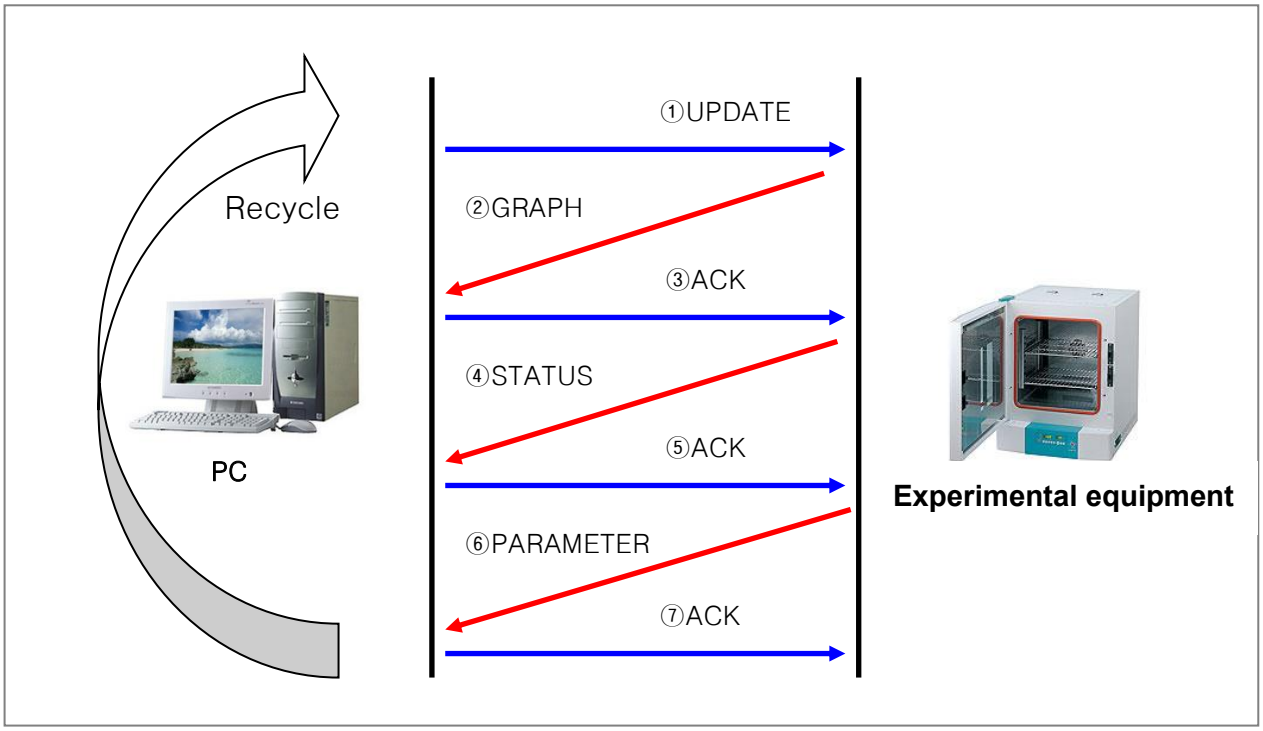
RiverLogics (주)리버로직스

Picture 1. Communication connect Diagrams

3. Protocol flowchart

LabTracer is Master which is leading communication and transport Control Packet. And Experimental equipment is Slave which is responding from Master Control Packet. There is UPDATE Packet to issue Control Packet by Master. The Master sends UPDATE Packet and demand to Slave (experiment equipment). So, experiment equipment information can be monitoring.

Slave as each equipment need to transmit GRAPH Packet, STATUS Packet, PARAMETER Packet in sequence to Master as Lab Tracer who needs to transmit ACK Packet in response to Slave's request.. Lab Tracer's one cycle of UPDATE Packet consists of these packets and ACK Packets in response to last Parameter Packet. Lab Tracer will start again for the next cycle of protocol by transmitting UPDATE Packet. Like this way, Master's UPDATE Packet can be repeated continuously, and also can be monitored about information of each equipment.



Picture2**. Update cycle**

GRAPH Packet provides the information about PV value and heating volume of Temp. sensor. This information consists of 2 byte compressed, not by 4 byte, which will be useful for the users who need to store the data of Graph for a long time. STATUS Packet has information of the laboratory instrument's state and each one bite to show run and over temp. PARAMETER Packet represents Temp. Sensor PV, SV, Heat,

Run time. These values which are composed of more precise 4 bite can substitute for information of graph packet. Lab Tracer also can inform the instrument of Run and Stop by using ACK Packet. The experimenters user can use unique operation program which developed by user with utilizing above packets.

This lab tracer's protocol can be used into the following functions through the software applied by the users' own developed

- SV Setting : SV can be transmitted, or revised to each equipment in the software applied.
- PV detecting : PV can be monitored from the each equipment in the software applied.
- Heating volume detecting : Heating volume can be monitored from the each equipment in the software applied.
- " Run " checking : Run status can be monitored from the each equipment in the software applied.
- " Over Temp " checking : OverTemp status can be monitored from the each equipment in the software applied.
- " Run/Stop " checking : Run/Stop command can be done to each equipment in the software applied.
- " Model No " checking: Model No. can be monitored from the each equipment in the software applied.
- " Run Time " checking : Run Time can be monitor from the each equipment when in operation in the software applied..

4. Packet

Table 1 and 2 shows packet that LabTracer and equipment trnsmit. Internal packet is divided 6 part and these are Header, Data Length, Extension, Command, Data, and Checksum.

4.1 Header

This is start field of packet. Table 1 shows Header of packet which master transmit. Table 2 shows Header of packet which slave send. It is 4Byte and Hex Code. Each represents 0x01AA00AA and 0x00AA01AA.

4.2 Data Length

It consists of Hex Code of 1 byte and represents length of data. The length of data can be changeable. So, total length of packet can be recognized by this code.

4.3 Extension

Extension consists of Hex code of 2 byte. A part of bit among 16 bit is used to shows status of equipment or LabTracer or performance of function. The rest of bit can be used for further additional function of protocol. This extension can be set and transmitted with only LabTracer and the equipment which is transmitted verifies setting value and applies to equipment.

(For example)

- LabTracer change set value (SV) and transmit it to equipment : 0002 (hex)
- LabTracer starts or stops equipment : 0001 (hex)

4.4 Command

Command consists of Hex code of 1 byte and shows function of each packet. There are 5 different kind of Command and they are UPDATE, GRAPH, STATUS, PARAMETER, and ACK.

- UPDATE : LabTracer needs information of equipment. 30(hex) occurrence
- GRAPH : Equipment transmit GRAPH information to LabTracer. 31(hex) occurrence
- STATUS : Equipment transmit STATUS information to LabTracer. 32(hex) occurrence
- PARAMETER: Equipment transmit PARAMETER information to LabTracer. 33(hex) occurrence
- ACK : LabTracer has information on equipment and transmits receipt confirmation packet. 01 (hex) occurrence

4.5 Data

Data consists of information of each packet and shows set temperature, temperature sensor output, quantity of heating, equipment status, and etc. Data consist of Hex code and its length is stored in Data size. Detail information on Data will be next chapter.

4.6 Checksum

Checksum consists of hex code of 2 byte and it is used for sensing possible transmittance error during packet transmittance. It shows Arithmetic Sum of all byte of Header, Data Length, Extension, Command, and Data. It consists of 2 byte and it is total sum from Header to Data sequentially. The following is example of receiving confirmation packet.

Ex) **[01 AA 00 AA 01 00 00 01 FF 02 56]**

01 + AA = AB,
AB + 00 = AB
AB + AA = 155
155 + 01 = 156
156 + 00 = 156
156 + 00 = 156
156 + 01 = 157
157 + FF = 256 (Checksum of this packet is 0256.)

- Hex Code -

Header	Data Length	Extension	Command	Data	Checksum
01 AA 00 AA	??	?? ??	??	?? ??	?? ??
(4 byte)	(1 byte)	(2 byte)	(1 byte)	(Variable)	(2 byte)

Table 1. Master Packet

- Hex Code -

Header	Data Length	Extension	Command	Data	Checksum
00 AA 01 AA (4 byte)	?? (1 byte)	?? ?? (2 byte)	?? (1 byte)	?? ?? (Variable)	?? ?? (2 byte)

Table 2. Slave Packet

5. Protocol Streams

It checks detail content of Protocol streams and each dates with giving an example which is transmitter-receiver packet between LabTracer and laboratory equipments. All packet of LabTracer and current laboratory equipment are composed of information on the hexadecimal basis.

Following protocol shows one cycle from starting UPDATE Packet to transmitting last ACK Packet.

5.1 UPDATE (PC → Controller)

[01 AA 00 AA 04 00 00 30 FF 00 00 00 02 88]

1. 01 AA 00 AA – Header
2. 04 – Data Length
3. 00 00 – Extension
4. 30 – Command(Update)
5. FF – Data(Reserved)
6. 00 00 – Data (SV set) : Signed Integer (2's Complement)
   7FFF(hex) → 32767(deci) → +327.67°C
   8000(hex) → -32768(deci) → -327.68°C
7. 00 – Data(Reserved)
8. 02 88- Checksum

- Hex Code -

Header	Data Length	Extension (SV 100°C set)	Command	Data			Checksum
				Reserved	SV Value	Reserved	
01AA00AA	04	0000 (0002)	30	FF	0000 (2701)	00	0288 (02B2)

5.2 GRAPH (Controller → PC)

[00 AA 01 AA 2A F4 61 31 F9 63 92 FF C0 02 80 FF FC 00 64 00 29 00 00 27 10 00 00 00 00 01 27 10 00 00 00 00 02 00 00 00 00 00 00 03 00 01 00 00 00 00 0A 31]

1. 00 AA 01 AA – Header(Controller -> PC)
2. 2A – Data Length
3. F4 61 – Extension
4. 31 – Command(Graph Value)
5. F9 63 92 FF C0 02 80 FF – Data(PVFND, SVFND)
6. FC – Data(LED)

7. 00 - Data(Reserved)
8. 64 – Data(Buzzer)
9. 00 - Data(Reserved)
10. 29 – Data(Model NO) : 41(Deci) represents OV-11 model (see model table)
11. 00 – Data(Reserved)
12. 00 27 10 00 00 00 00 – PV Data(Instance, Value, Reserved)

    Instance 00 : the number in graph packet indicates PV Date

    Value 2710 : Signed Integer (2's Complement)

    2710(hex) → 10000(deci) → +100.00°C

    7FFF(hex) → 32767(deci) → +327.67°C

    8000(hex) → -32768(deci) → -327.68°C

    Reserved 00000000 : Don't care
13. 01 27 10 00 00 00 00 – Heat Data(Instance, Value, Reserved)

    Instance 01 : the number in graph packet indicates Heat Data

    Value 2710 : Signed Integer (2's Complement)

    2710(hex) → 10000(deci) → 100.00%

    Reserved 00000000 : Don't care
14. 02 00 00 00 00 00 00 - Data(Reserved)
15. 03 00 01 00 00 00 00 - Data(Reserved)
16. 0A 31 – Checksum

- Hex Code -

Header	Data Length	Extension (Don't care)	Command	Basic Data		
				Don't care	Model No	Reserved
00AA01AA	2A	F461	31	F9 ~ 00	29	00

PV Data			Heat Data		
Instance	Value	Reserved	Instance	Value	Reserved
00	2710	00000000	01	2710	00000000

Reserved		Reserved		Checksum
Instance	Reserved	Instance	Reserved	
02	000000000000	03	000000000000	0A31

5.3 ACK (PC → Controller)

[01 AA 00 AA 01 00 01 01 FD 02 55]

9. 01 AA 00 AA – Header(PC -> Controller)
10. 01 – Data Length
11. 00 01 – Extension

    00 00 : in case not informing Run or Stop(no Run change)

    00 01 : in case of informing Run or Stop

12. 01 – Command(ACK)
13. FD – Data

    FF : no Run

    FD : informing Run or Stop
14. 02 55- Checksum

- Hex Code -

Header	Data Length	Extension (no Run change)	Command	Data (no Run change)	Checksum
01AA00AA	01	0001 (0000)	01	FD (FF)	0255 (0256)

5.4 STATUS (Controller → PC)

[00 AA 01 AA 1A F4 61 32 A4 63 92 FF C0 19 C0 F9 FC 00 64 00 29 00 04 01 05 00 06 00 07 00 08 00 09 00 09 D1]

1. 00 AA 01 AA – Header(Controller -> PC)
2. 1A – Data Length
3. F4 61 – Extension
4. 32 – Command(Level Value)
5. A4 63 92 FF C0 19 C0 F9 – Data(PVFND, SVFND)
6. FC – Data(LED)
7. 00 - Data(Reserved)
8. 64 – Data(Buzzer)
9. 00 - Data(Reserved)
10. 29 – Data(Model NO) : 41(Deci) represents OV-11 model (see model table)
11. 00 – Data(Reserved)
12. 04 01 – Run Data(Instance, Status Value)

    Instance 04 : number in status packet indicates Run Date

    Status Value 01 : represent the status of laboratory unit's operation

    Status Value 00 : represent the status of not operating
13. 05 00 – Data(AutoTune-Instance, AutoTune -Status)
14. 06 00 - Data(ProgramRun-Instance, ProgramRun -Status)
15. 07 00 - OverTemp Data(Instance, Status Value)

    Instance 07 : the number in status packet indicates OverTemp Data

    Status Value 01 : represent over temperature or abnormal temperature

    Status Value 00 : represent the unit normal
16. 08 00 - Data(WaterLevel-Instance, WaterLevel -Status)
17. 09 00 - Data(Cooler-Instance, Cooler -Status)
18. 09 D1 – Checksum

- Hex Code -

Header	Data Length	Extension (Don't care)	Command	Basic Data		
				Don't care	Model No	Reserved
00AA01AA	1A	F461	32	A4~00	29	00

- Hex Code -

Run Data		Don't care			
Instance	Value	Instance	Value	Instance	Value
04	01	05	00	06	00

- Hex Code -

Over temp Data		Don't care				Checksum
Instance	Value	Instance	Value	Instance	Value	
07	00	08	00	09	00	09 D1

5.5 ACK (PC → Controller)

[01 AA 00 AA 01 00 01 01 FD 02 55]

15. 01 AA 00 AA – Header(PC -> Controller)
16. 01 – Data Length
17. 00 01 – Extension

    00 00 : in case of not informing Run or Stop(no Run change)

    00 01 : in case of informing Run or Stop
18. 01 – Command(ACK)
19. FD – Data

    FF : no Run

    FD : informing Run or Stop
20. 02 55- Checksum

- Hex Code -

Header	Data Length	Extension (no Run change)	Command	Data (no Run change)	Checksum
01AA00AA	01	0001 (0000)	01	FD (FF)	0255 (0256)

5.6 PARAMETER (Controller → PC)

[00 AA 01 AA 40 F4 E1 33 B0 63 92 9C C0 24 A4 F9 FC 00 64 00 29 00 0A 00 00 27 10 0B 00 00 75 3A 0C 00 00 27 10 0D 00 00 08 E0 0E 00 00 00 00 0F 00 00 00 00 10 00 00 17 6F 11 00 00 00 00 12 00 00 00 00 13 00 00 00 00 0D 04]

1. 00 AA 01 AA – Header(Controller -> PC)
2. 40 – Data Length
3. F4 E1 – Extension
4. 33 – Command(Level Value)
5. B0 63 92 9C C0 24 A4 F9 – Data(PVFND, SVFND)
6. FC – Data(LED)
7. 00 - Data(Reserved)
8. 64 – Data(Buzzer)
9. 00 - Data(Reserved)
10. 29 – Data(Model NO) : 41(Deci) represents OV-11 model (refer to model table)
11. 00 – Data(Reserved)
12. 0A 00 00 27 10 – PV Data(Instance, Value)

    Instance 0A : number which represent PV Data in Parameter packet

    Value : Signed Integer (2's Complement)

    00002710(hex) → 10000(deci) → +100.00°C

    00007FFF(hex) → 32767(deci) → +327.67°C

    FFFF8000(hex) → -32768(deci) → -327.68°C
13. 0B 00 00 75 3A – SV Data(Instance, Value)

    Instance 0B : number which represent SV Data in Parameter packet

    Value : Signed Integer (2's Complement)

    00002710(hex) → 10000(deci) → +100.00°C

    00007FFF(hex) → 32767(deci) → +327.67°C

    FFFF8000(hex) → -32768(deci) → -327.68°C
14. 0C 00 00 27 10 – Heat Data(Instance, Value)

    Instance 0C : number which represent Heat Data in Parameter packet

    Value : Signed Integer (2's Complement)

    00002710(hex) → 10000(deci) → 100.00%
15. 0D 00 00 08 E0 - RunTime Data(Instance, Value)

    Instance 0D : number which represent Run Time Data in Parameter packet

    Value : 000008E0(unit is Second)

000008E0 = 2272(deci) → 37 Min : 52 Sec

16. 0E 00 00 00 00 - Data(OnTime-Instance, OnTimeCount)
17. 0F 00 00 00 00 - Data(OffTime-Instance, OffTimeCount)
18. 10 00 00 17 6F – Data(Power-Instance, PowerFrequency)
19. 11 00 00 00 00 – Data(Reserved-Instance, reserved)
20. 12 00 00 00 00 – Data(Reserved-Instance, reserved)
21. 13 00 00 00 00 – Data(Reserved-Instance, reserved)
22. 0D 04 - Checksum

- Hex Code -

Header	Data Length	Extension	Command	Basic Data		
				Don't care	Model No	Reserved
00AA01AA (4 byte)	1A (1 byte)	F4E1	33 (1 byte)	B0~00 (12 byte)	29 (1 byte)	00 (1 byte)

- Hex Code -

PV Data		SV Data	
Instance	Value	Instance	Value
0A	00002710	0B	0000753A

- Hex Code -

Heat Data		Run Time Data	
Instance	Value	Instance	Value
0C	00002710	0D	000008E0

- Hex Code -

Don't care		Don't care	
Instance	Value	Instance	Value
0E	00000000	0F	00000000

- Hex Code -

Don't care		Reserved	
Instance	Value	Instance	Reserved
10	0000176F	11	000000

- Hex Code -

Reserved		Reserved		Checksum
Instance	Reserved	Instance	Reserved	

12	000000	13	000000	0D04

5.7 ACK (PC → Controller)

[01 AA 00 AA 01 00 01 01 FD 02 55]

21. 01 AA 00 AA – Header(PC -> Controller)
22. 01 – Data Length
23. 00 01 – Extension

    00 00 : in case not informing Run or Stop(no Run change)

    00 01 : in case informing Run or Stop
24. 01 – Command(ACK)
25. FD – Data

    FF : no Run

    FD : informing Run or Stop
26. 02 55- Checksum

- Hex Code -

Header	Data Length	Extension (no Run change)	Command	Data (no Run change)	Checksum
01AA00AA	01	0001 (0000)	01	FD (FF)	0255 (0256)

6. Model NO

Below table shows the Model No. of JeioTech's products which can communicate.

table1. Model No.

Model No (Decimal)	Description	Model Name
01	Oven Forced convection Small	OF-02G
02	Oven Forced convection Medium	OF-12G
03	Oven Forced convection Large	OF-22G
21	Oven Natural convection Small	ON-02G
22	Oven Natural convection Medium	ON-12G
23	Oven Natural convection Large	ON-22G
41	Oven Vacuum Small 220V	OV-11
42	Oven Vacuum Large 220V	OV-12
51	Oven Vacuum Small 100V	VO-300

52	Oven Vacuum Large 100V	VO-400
61	Incubator Small	IB-05G
62	Incubator Medium	IB-15G
63	Incubator Large	IB-25G
81	Incubator Ref. Small	IL-11
82	Incubator Ref. Large	IL-21
101	Bath Water Small	BW-05G
102	Bath Water Medium	BW-10G
103	Bath Water Large	BW-20G
121	Circulation Water Small	CW-05G
122	Circulation Water Medium	CW-10G
123	Circulation Water Large	CW-20G
141	Ref. Water	RW-0525G
142	Ref. Water	RW-1025G

Model No (Decimal)	Description	Model Name
143	Ref. Water	RW-2025G
144	Ref. Water	RW-3025G
145	Ref. Water	RW-0540G
146	Ref. Water	RW-1040G
147	Ref. Water	RW-2040G
148	Ref. Water	RW-3040G
161	Visual Bath	VB-40G
162	Visual Bath	VB-55G
181	Shaking Incubator	IS-350
182	Shaking Incubator	IS-350R
183	Shaking Incubator	IS-650
184	Shaking Incubator	IS-65./0R
185	Shaking Incubator	IS-971
186	Shaking Incubator	IS-971R
187	Shaking Incubator	IS-971RF
201	Personal Incubator	IL-P

7. Program example

Simple example program was made by using visual C++ above explained. We'd like to increase to utilize main protocol by attaching the code with explanation.